

Technical Report – CAGS Project: Autonomous Detection and Repair System (Project ARES)

1. General Objective

The main goal of this project was to create, train, and design an Artificial Intelligence (AI) system for the drones of **CAGS – Project ARES**, aimed at detecting and assisting in the repair of damages in space structures, specifically rockets and the **International Space Station (ISS)**.

The system focuses on the development of two types of drones named **CAGS_F413**, where **CAGS** represents the initials of the creators (Camilo, Ángel, Gustavo, and Sara), and **F413** refers to Philippians 4:13 – *“I can do all things through Christ who strengthens me.”*

2. Project Description

A dual prototype was designed and tested to represent both drones of the system:

1. **Scanning Drone (CAGS_F413-S):**

Equipped with computer vision and AI, this drone can analyze space structures and detect visible or potential damage on surfaces such as solar panels, energy modules, or thermal shields.

2. **Repair Drone (CAGS_F413-R):**

Responsible for carrying components and tools to physically repair the affected areas. Although only one prototype was built, it integrates both roles — detection and response.

3. AI Training

The AI was trained using **TensorFlow** and **Keras**, based on the **VGG16** architecture pre-trained with ImageNet.

Training was carried out in multiple stages using datasets of approximately **3,000 images per session**, generated from **3D simulations in Blender**.

Data Source

- A 3D model of the **International Space Station (ISS)** was obtained from the official **NASA** website.

- In **Blender**, various types of damage were simulated on the **solar arrays**, including cracks, dirt, and microfractures.
- Images were automatically exported through **Python scripts**, producing a diverse and balanced dataset for model training.

4. Training Results

Several model versions were tested, each with different hyperparameters and image augmentations.

- **First training:** Accuracy $\approx$ 50% (limited and unbalanced dataset)
- **Second training:** Accuracy $\approx$ 70% (improved variability in lighting and angles)
- **Third training:** Accuracy $\approx$ 91% (stable model with data augmentation)
- **Final training:** Accuracy **99.27%** on the test set

The final model was obtained after **12 epochs**, using **Early Stopping** to prevent overfitting. These results demonstrate progressive improvement until achieving a robust and highly accurate AI capable of detecting damage with near-perfect reliability.

5. Technical Implementation

- **Programming language:** Python
- **Framework:** Flask (web interface and local API)
- **Libraries:** TensorFlow, Keras, OpenCV, NumPy, PIL
- **Simulation engine:** Blender
- **Training environment:** Google Colab (Tesla T4 GPU)
- **Base architecture:** VGG16 (with customized dense layers and Dropout regularization)

The developed web application allows the user to:

- Upload an image of a solar panel.
- Process it using the trained AI.
- Display:

- Damage detection results
- Confidence level
- Estimated damaged area percentage
- Suggested repair action

6. Repair Simulation

Beyond detection, a **repair simulation module** was integrated.

Once a damage is detected, the AI autonomously selects a response action from the following options:

- **clean**
- **patch**
- **replace**
- **tighten_screws**

This function mimics the repair drone's behavior, displaying the decision with a short delay to simulate a real-time evaluation and repair process.

7. Conclusions

Despite having only **one and a half days of development**, the team achieved remarkable results:

- An AI model with **99.27% detection accuracy**.
- A functional and interactive **web environment**.
- A system capable of **recognizing, analyzing, and responding** to damage autonomously.

These results mark the **first functional version** of the AI system for **CAGS – Project ARES**, paving the way toward **autonomous space maintenance** through intelligent drones.

8. Author

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CAGS – Project ARES

“Autonomous Space Exploration and Repair”